

MOSAM SANJAY BISWAS

Software Engineer | AI/ML and Full-Stack Development | Dombivli, Mumbai, India

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Summary

2026 B.Tech graduate in AI and Data Science who builds AI systems and ships them as working software: an AI form-filling Chrome extension, a fully local voice assistant (29K lines of Python, 849 tests), a mental health platform used by 100+ students, and offline apps for two small businesses. Software internship with client delivery; co-authored IEEE-published ML research.

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, Dart, SQL, C++, C, HTML/CSS

AI/ML: PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, SHAP, OpenCV, LLMs (OpenAI, Gemini, Claude, Ollama)

Backend & Web: FastAPI, Flask, Node.js, Express.js, Next.js, React, Angular, REST APIs, WebSockets, Chrome Extensions

Mobile & Databases: Flutter, Android (Java), Firebase, MySQL, PostgreSQL, MongoDB, SQLite

Tools & Practices: Git, GitHub Actions (CI/CD), AWS, Pandas, Postman, Google OR-Tools, Unit & E2E Testing, Agile/Scrum

Experience

Sedna Technologies

Jan 2026 – Apr 2026

Software Developer Intern

Mumbai, India

- Shipped work across **4 client projects**, working directly with clients to gather requirements and resolve issues.
- Designed and built a **client-facing dashboard solo** with **Angular, Flask, and MySQL** over **REST APIs**, owning it from requirements to handover (later submitted as B.Tech dissertation).

Projects

Fill.ai ([install](#)) | *JavaScript, Chrome Extension (MV3), Ollama, Gemini API, Claude API, pdf.js*

2026

- Built and released a **Chrome extension that fills web forms with AI**: it turns a resume into an editable profile and fills job applications via Ollama, Gemini, or Claude behind **one shared JSON schema**.
- Blocked hallucinated answers in code, not prompts: every answer must **cite a profile field** or it is discarded, emails, phones, and dropdown options are verified, and sensitive fields, consent boxes, and passwords are never auto-filled.
- Handled React-controlled inputs, shadow DOM, and cross-origin iframes with read-back checks; shipped via a **GitHub Actions CI/CD** release pipeline with a **92-check end-to-end suite**, tested on live Greenhouse and Lever forms.

AMINAL | *Python, Whisper, Ollama (Llama 3.2), Qwen2.5-VL, Piper, OpenCV, SQLite*

Jul 2026 – Present

- Built a **local-first voice assistant** for Windows running speech-to-text, LLM intent parsing, vision, and text-to-speech **fully on-device** on an RTX 4060, with no cloud services in the voice path.
- Grew it to **29K lines of Python across 72 modules** with **125 voice commands** and **849 regression tests**, plus LLM output guards and window control by element name via **Windows UI Automation**.

MindFirst | *Python, Flask, Next.js, Prisma, NextAuth, Tailwind CSS*

2025

- Developed a **mental health pre-screening platform used by 100+ students**, pairing a chatbot with PHQ-9/GAD-7 psychometric screening at **95% accuracy**; its triage model comes from the IEEE paper below.
- Integrated **NextAuth** authentication and personalized mental health reports, **cutting user drop-off by 30%**.

IntelliStatement | *Python, FastAPI, OpenAI API, Streamlit*

2025

- Built an **LLM-powered bank statement analyzer**: **FastAPI** endpoints take PDF and image statements, run **LLM-based transaction extraction** into structured outputs, and feed a Streamlit dashboard of categorized spending insights.

PixelForge | *Python, PyTorch, CNNs, GANs*

2025

- Trained a CNN for **leukemia cell classification** (C-NMC) at **0.90 ROC-AUC** and **95% recall**, and built a custom **GAN** with a spectral-norm discriminator to synthesize cell images for scarce-data augmentation.

Pops & Order Control | *Flutter, Dart, SQLite, Android*

2026

- Built two **offline Android apps for real small businesses**, a fabrication firm's order tracker (replacing its paper diary) and a food stall order board, on **relational SQLite schemas** with cascades, migrations, and unit tests.

Research & Publications

- Comparative Analysis of Psychometric Data for Assessing Mental Health Testing Needs Using Machine Learning Models**, co-author. **IEEE Xplore**, Proc. ICAIHI 2025. DOI: [10.1109/ICAIIHI67124.2025.11403838](https://doi.org/10.1109/ICAIIHI67124.2025.11403838)
- Benchmarked LR, RF, SVM, XGBoost, and LightGBM on **28.5K PHQ-9/GAD-7 records** (stratified 5-fold CV, paired t-tests); LR led at **99% accuracy** with **SHAP** explanations, and the pre-screening flow roughly halves screening overhead.

Education

Ramrao Adik Institute of Technology, D. Y. Patil Deemed to be University

2022 – 2026

B.Tech in Computer Engineering (Major: Artificial Intelligence and Data Science), CGPA: 7.5

Navi Mumbai, India

Certifications, Leadership & Community

Certifications: Samsung Innovation Campus: AI | AWS Academy Graduate: Machine Learning | [Generative AI \(edX\)](#) | Agile & Scrum Fundamentals | Android Compose Camp (GDSC) | Industrial Approach to Java (CSI)

Leadership: Technical Head, Photocircle RAIT (2024–25): 3 workshops for 75+ students, **participation up 60%**

Community: Google Developer Groups (**25+ codelabs**, Google Cloud Innovator) | Hacktoberfest | SIH 2025 internal round